TUTORIAL II

THE DOCUMENT

II.1. DOCUMENT CLASS

We now describe how an entire document with chapters and sections and other embellishments can be produced with LaTeX. We have seen that all LaTeX files should begin by specifying the kind of document to be produced, using the command `\documentclass{... }`. We've also noted that for a short article (which can actually turn out to be quite long!) we write `\documentclass{article}` and for books, we write `\documentclass{book}`. There are other *document classes* available in LaTeX such as `report` and `letter`. All of them share some common features and there are features specific to each.

In addition to specifying the type of document (which we *must* do, since LaTeX has no default document class), we *can* also specify some options which modify the default format.Thus the actual syntax of the `\documentclass` command is

`\documentclass[`*options*`]{`*class*`}`

Note that options are given in *square brackets* and not braces. (This is often the case with LaTeX commands—options are specified within square brackets, after which mandatory arguments are given within braces.)

II.1.1. Font size

We can select the size of the font for the normal text in the entire document with one of the options

`10pt` `11pt` `12pt`

Thus we can say

```
\documentclass[11pt]{article}
```

to set the normal text in our document in 11 pt size. The default is `10pt` and so this is the size we get, if we do not specify any font-size option.

II.1.2. Paper size

We know that LaTeX has its own method of breaking lines to make paragraphs. It also has methods to make vertical breaks to produce different pages of output. For these breaks to work properly, it must know the width and height of the paper used. The various options for selecting the paper size are given below:

`letterpaper`	11×8.5 in	`a4paper`	20.7×21 in
`legalpaper`	14×8.5 in	`a5paper`	21×14.8 in
`executivepaper`	10.5×7.25 in	`b5paper`	25×17.6 in

Normally, the longer dimension is the vertical one—that is, the height of the page. The default is `letterpaper`.

II.1.3. Page formats

There are options for setting the contents of each page in a single column (as is usual) or in two columns (as in most dictionaries). This is set by the options

```
onecolumn    twocolumn
```

and the default is `onecolumn`.

There is also an option to specify whether the document will be finally printed on just one side of each paper or on both sides. The names of the options are

```
oneside    twoside
```

One of the differences is that with the `twoside` option, page numbers are printed on the right on odd-numbered pages and on the left on even numbered pages, so that when these printed back to back, the numbers are always on the outside, for better visibility. (Note that LaTeX has no control over the actual *printing*. It only makes the *formats* for different types of printing.) The default is `oneside` for `article`, `report` and `letter` and `twoside` for `book`.

In the `report` and `book` class there is a provision to specify the different chapters (we will soon see how). Chapters always begin on a new page, leaving blank space in the previous page, if necessary. With the `book` class there is the additional restriction that chapters begin only on odd-numbered pages, leaving an entire page blank, if need be. Such behavior is controlled by the options,

```
openany    openright
```

The default is `openany` for `reportclass` (so that chapters begin on "any" *new* page) and `openright` for the `book` class (so that chapters begin only on *new* `right`, that is, odd numbered, page).

There is also a provision in LaTeX for formatting the "title" (the name of the document, author(s) and so on) of a document with special typographic consideration. In the `article` class, this part of the document is printed along with the text following on the first page, while for `report` and `book`, a *separate* title page is printed. These are set by the options

```
notitlepage    titlepage
```

As noted above, the default is `notitlepage` for `article` and `titlepage` for `report` and `book`. As with the other options, the default behavior can be overruled by explicitly specifying an option with the `documentclass` command.

There are some other options to the `documentclass` which we will discuss in the relevant context.

II.2. PAGE STYLE

Having decided on the overall appearance of the document through the `\documentclass` command with its various options, we next see how we can set the style for the individual pages. In LaTeX parlance, each page has a "head" and "foot" usually containing such information as the current page number or the current chapter or section. Just what goes where is set by the command

```
\pagestyle{...}
```

where the mandatory argument can be any one of the following *styles*

```
plain    empty    headings    myheadings
```

The behavior pertaining to each of these is given below:

plain The page head is empty and the foot contains just the page number, centered with respect to the width of the text. This is the default for the `article` class if no `\pagestyle` is specified in the preamble.

empty Both the head and foot are empty. In particular, no page numbers are printed.

headings This is the default for the `book` class. The foot is empty and the head contains the page number and names of the chapter section or subsection, depending on the document class and its options as given below:

CLASS	OPTION	LEFT PAGE	RIGHT PAGE
`book, report`	one-sided	—	*chapter*
	two-sided	*chapter*	*section*
`article`	one-sided	—	*section*
	two-sided	*section*	*subsection*

myheadings The same as `headings`, except that the 'section' information in the head are not predetermined, but to be given explicitly using the commands `\markright` or `\markboth` as described below.

Moreover, we can customize the style for the *current page* only using the command

`\thispagestyle{`*style*`}`

where *style* is the name of one of the styles above. For example, the page number may be suppressed for the current page alone by the command `\thispagestyle{empty}`. Note that only the *printing* of the page number is suppressed. The next page will be numbered with the next number and so on.

II.2.1. Heading declarations

As we mentioned above, in the page style `myheadings`, we have to specify the text to appear on the head of every page. It is done with one of the commands

`\markboth{`*left head*`}{`*right head*`}`
`\markright{`*right head*`}`

where *left head* is the text to appear in the head on left-hand pages and *right head* is the text to appear on the right-hand pages.

The `\markboth` command is used with the `twoside` option with even numbered pages considered to be on the left and odd numbered pages on the right. With `oneside` option, all pages are considered to be right-handed and so in this case, the command `\markright` can be used. These commands can also be used to override the default head set by the `headings` style.

Note that these give only a limited control over the head and foot. since the general format, including the font used and the placement of the page number, is fixed by LaTeX. Better customization of the head and foot are offered by the package fancyhdr, which is included in most LaTeX distributions.

II.3. PAGE NUMBERING

The style of page numbers can be specified by the command

`\pagenumbering{...}`

The possible arguments to this command and the resulting style of the numbers are given below:

`arabic`	Indo-Arabic numerals
`roman`	lowercase Roman numerals
`Roman`	upper case Roman numerals
`alph`	lowercase English letters
`Alph`	uppercase English letters

The default value is `arabic`. This command resets the page *counter*. Thus for example, to number all the pages in the 'Preface' with lowercase Roman numerals and the rest of the document with Indo-Arabic numerals, declare `\pagenumbering{roman}` at the beginning of the Preface and issue the command `\pagestyle{arabic}` immediately after the first `\chapter` command. (The `\chapter{...}` command starts a new chapter. We will come to it soon.)

We can make the pages start with any number we want by the command

`\setcounter{page}{`*number*`}`

where *number* is the page number we wish the current page to have.

II.4. FORMATTING LENGTHS

Each page that LaTeX produces consists not only of a *head* and *foot* as discussed above but also a *body* (surprise!) containing the actual text. In formatting a page, LaTeX uses the width and heights of these parts of the page and various other lengths such as the left and right margins. The values of these lengths are set by the paper size options and the page format and style commands. For example, the page layout with values of these lengths for an odd page and even in this book are separately shown below.

These lengths can all be changed with the command `\setlength`. For example,

```
\setlength{\textwidth}{15cm}
```

makes the width of text 15 cm. The package geometry gives easier interfaces to customize page format.

II.5. PARTS OF A DOCUMENT

We now turn our attention to the contents of the document itself. Documents (especially longer ones) are divided into chapters, sections and so on. There may be a title part (sometimes even a separate title page) and an abstract. All these require special typographic considerations and LaTeX has a number of features which automate this task.

II.5.1. Title

The "title" part of a document usually consists of the name of the document, the name of author(s) and sometimes a date. To produce a title, we make use of the commands

`\title{`*document name*`}`
`\author{`*author names*`}`
`\date{`*date text*`}`

`\maketitle`

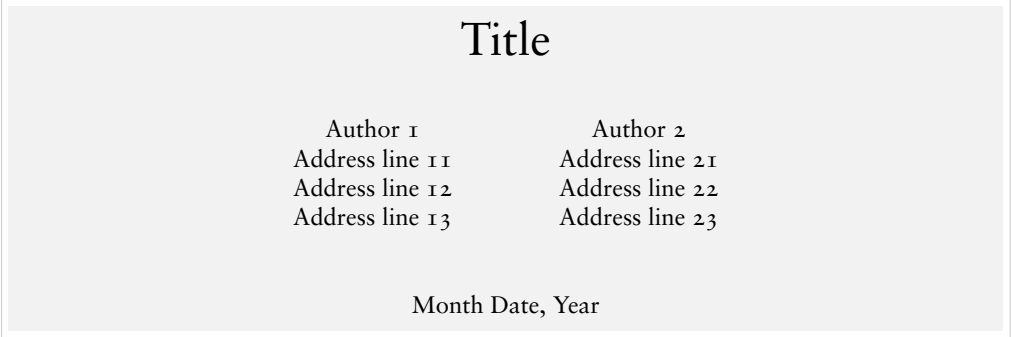
Note that after specifying the arguments of `\title`, `\author` and `\date`, we must issue the command `\maketitle` for this part to be typeset.

By default, all entries produced by these commands are centered on the lines in which they appear. If a title text is too long to fit in one line, it will be broken automatically. However, we can choose the break points with the `\\` command.

If there are several authors and their names are separated by the `\and` command, then the names appear side by side. Thus

```
\title{Title}
\author{Author 1\\
         Address line 11\\
         Address line 12\\
         Address line 13
         \and
         Author 2\\
         Address line 21\\
         Address line 22\\
         Address line 23}
 \date{Month Date, Year}
```

produces



If instead of `\and`, we use (plain old) `\\`, the names are printed one below another.

We may leave some of these arguments empty; for example, the command `\date{ }` prints no date. Note, however, that if you simply omit the `\date` command itself, the current date will be printed. The command

`\thanks{`*footnote text*`}`

can be given at any point within the `\title`, `\author` or `\date`. It puts a marker at this point and places the *footnote text* as a footnote. (The general method of producing a footnote is to type `\footnote{`*footnote text*`}` at the point we want to refer to.)

As mentioned earlier, the "title" is printed in a separate page for the document classes `book` and `report` and in the first page of the document for the class `article`. (Also recall that this behavior can be modified by the options `titlepage` or `notitlepage`.)

II.5.2. Abstract

In the document classes `article` and `report`, an abstract of the document in special format can be produced by the commands

```
\begin{abstract}        Abstract        Text
\end{abstract}
```

Note that we have to type the abstract ourselves. (There is a limit to what even LaTeX can do.) In the `report` class this appears on the separate title page and in the `article` class it appears below the title information on the first page (unless overridden by the title page option). This command is not available in the `book` class.

II.6. DIVIDING THE DOCUMENT

A book is usually divided into chapters and (if it is technical one), chapters are divided into sections, sections into subsections and so on. LaTeX provides the following hierarchy

of *sectioning* commands in the book and report class:

```
\chapter
\section
\subsection
\subsubsection
\paragraph
\subparagraph
```

Except for \chapter all these are available in article class also. For example, the heading at the beginning of this chapter was produced by

```
\chapter{The Document}
```

and the heading of this section was produced by

```
\section{Dividing the document}
```

To see the other commands in action, suppose at this point of text I type

```
\subsection{Example}
In this example, we show how subsections and subsubsections
are produced (there are no subsubsubsections). Note how the
subsections are numbered.

\subsubsection{Subexample}
Did you note that subsubsections are not numbered? This is so in the
\texttt{book} and \texttt{report} classes. In the \texttt{article}
class they too have numbers. (Can you figure out why?)

\paragraph{Note}
Paragraphs  and subparagraphs do not have numbers. And they have
\textit{run-in} headings.

Though named ``paragraph'' we can have several paragraphs of text
within this.

\subparagraph{Subnote}
Subparagraphs have an additional indentation too.

And they  can also contain more than one paragraph of text.
```

We get

II.6.1. Example

In this example, we show how subsections and subsubsections are produced (there are no subsubsubsections). Note how the subsections are numbered.

Subexample

Did you note that subsubsections are not numbered? This is so in the book and report classes. In the article class they too have numbers. (Can you figure out why?)

Note Paragraphs and subparagraphs do not have numbers. And they have *run-in* headings. Though named "paragraph" we can have several paragraphs of text within this.

Subnote Subparagraphs have an additional indentation too. And they can also contain more than one paragraph of text.

II.6.2. More on sectioning commands

In the `book` and the `report` classes, the `\chapter` command shifts to the beginning of a new page and prints the word "Chapter" and a number and beneath it, the name we have given in the argument of the command. The `\section` command produces two numbers (separated by a dot) indicating the chapter number and the section number followed by the name we have given. It does not produce any text like "Section". Subsections have three numbers indicating the chapter, section and subsection. Subsubsections and commands below it in the hierarchy do not have any numbers.

In the `article` class, `\section` is highest in the hierarchy and produces single number like `\chapter` in `book`. (It does not produce any text like "Section", though.) In this case, subsubsections also have numbers, but none below have numbers.

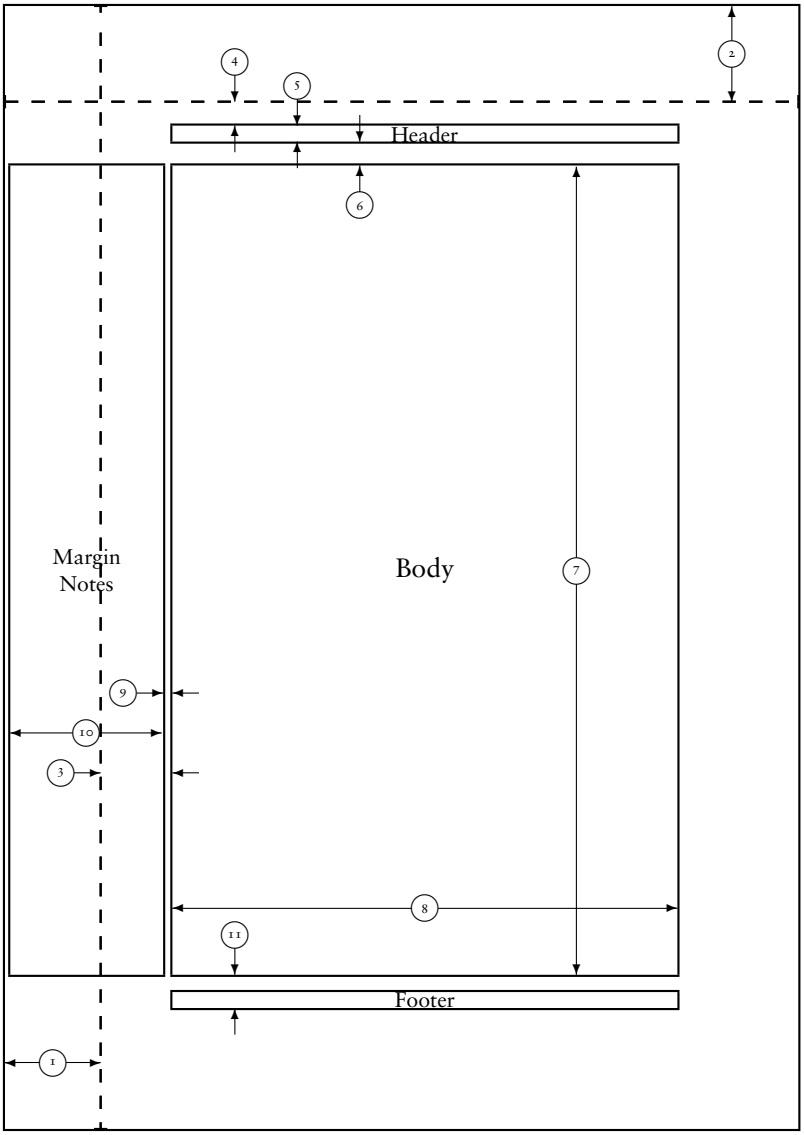
Each sectioning command also has a "starred" version which does not produce numbers. Thus `\section*{`*name*`}` has the same effect as `\section{`*name*`}`, but produces no number for this section.

Some books and longish documents are divided into parts also. LaTeX also has a `\part` command for such documents. In such cases, `\part` is the highest in the hierarchy, but it does not affect the numbering of the lesser sectioning commands.

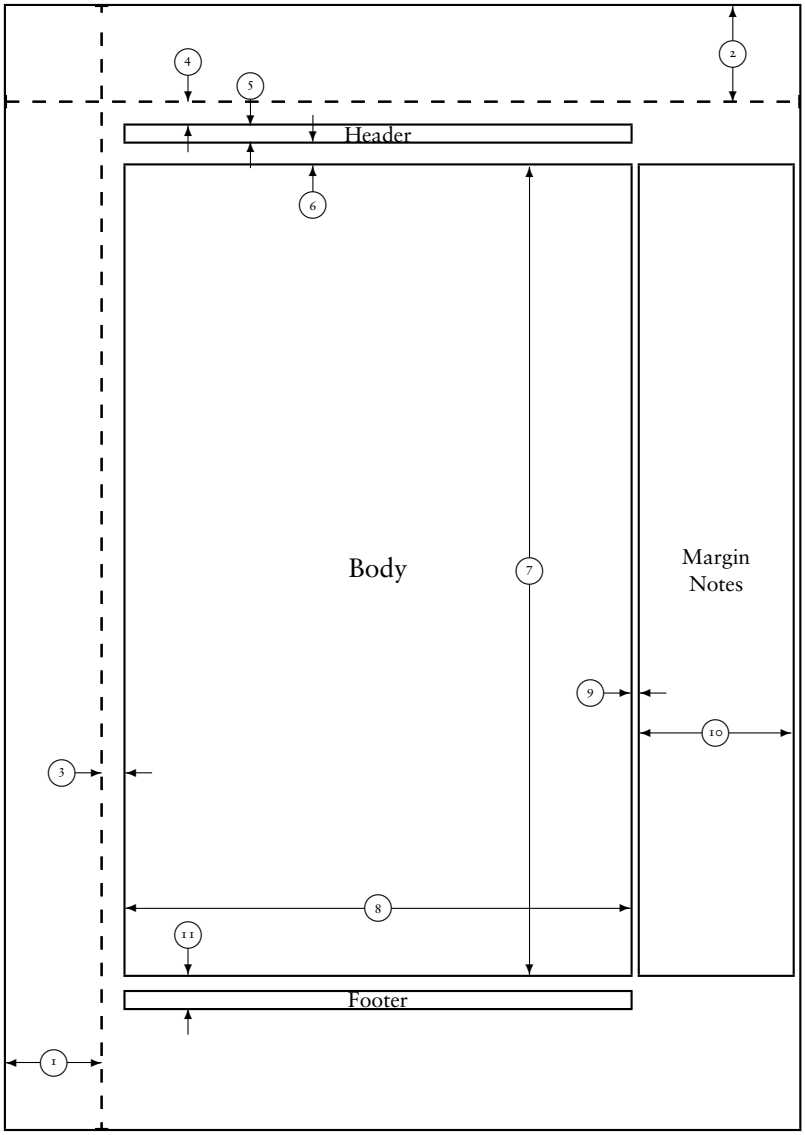
You may have noted that LaTeX has a specific format for typesetting the section headings, such as the font used, the positioning, the vertical space before and after the heading and so on. All these can be customized, but it requires some TeXpertise and cannot be addressed at this point. However, the package sectsty provided some easy interfaces for tweaking some of these settings.

II.7. WHAT NEXT?

The task of learning to create a document in LaTeX is far from over. There are other things to do such as producing a bibliography and a method to refer to it and also at the end of it all to produce a table of contents and perhaps an index. All these can be done efficiently (and painlessly) in LaTeX, but they are matters for other chapters.



1	one inch + \hoffset	2	one inch + \voffset
3	\evensidemargin = 54pt	4	\topmargin = 18pt
5	\headheight = 12pt	6	\headsep = 18pt
7	\textheight = 609pt	8	\textwidth = 380pt
9	\marginparsep = 7pt	10	\marginparwidth = 115pt
11	\footskip = 25pt		\marginparpush = 5pt (not shown)
	\hoffset = 0pt		\voffset = 0pt
	\paperwidth = 597pt		\paperheight = 845pt



1	one inch + \hoffset	2	one inch + \voffset
3	\oddsidemargin = 18pt	4	\topmargin = 18pt
5	\headheight = 12pt	6	\headsep = 18pt
7	\textheight = 609pt	8	\textwidth = 380pt
9	\marginparsep = 7pt	10	\marginparwidth = 115pt
11	\footskip = 25pt		\marginparpush = 5pt (not shown)
	\hoffset = 0pt		\voffset = 0pt
	\paperwidth = 597pt		\paperheight = 845pt